

Trainings ▾

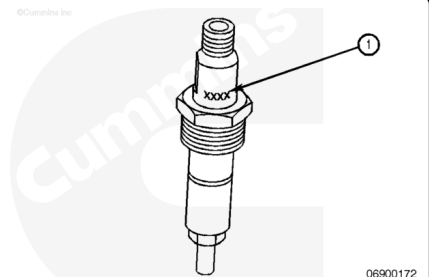
General Information

General Information

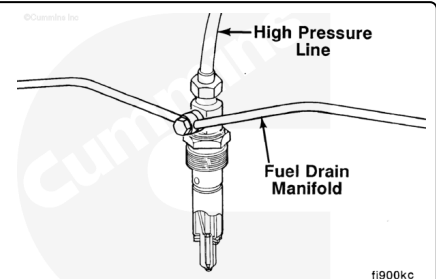
⚠ CAUTION ⚠

Use only the specified injector for the engine.

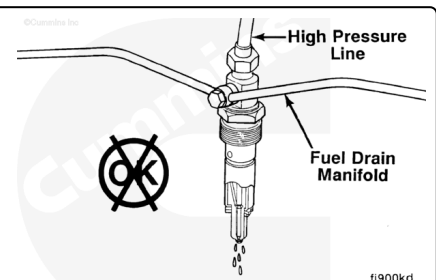
All engines use closed nozzle, hole-type injectors. However, the injectors can have different part numbers for different engine ratings. The last four digits of the Cummins part number are used to identify the injectors.



During the injection cycle, high pressure from the injection pump rises to the operating (pop) pressure, which causes the needle valve in the injector to lift. Fuel is then injected into the cylinder. A shimmed spring is used to force the needle valve closed as the injection pressure drops below the pop pressure to seal off the nozzle after injection.

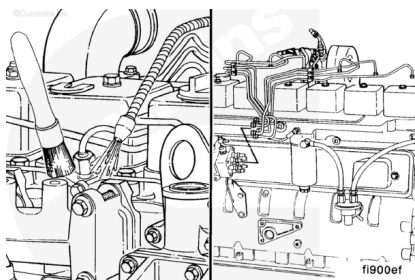


Failure of the needle valve to lift and close at the correct time or needle valve stuck open can cause the engine to misfire and produce low power. Fuel leaking from the open nozzle can cause a fuel knock, poor performance, smoke, poor fuel economy, and rough running.



Preparatory Steps

- Clean around the injectors.
- Disconnect the high-pressure fuel supply lines.
- Disconnect the fuel drain manifold.



Remove

Rust-Penetrating Solvent

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

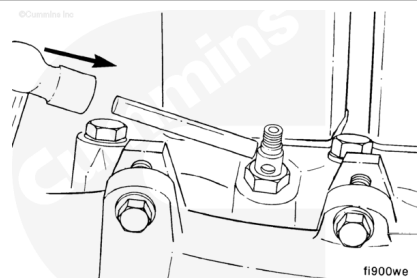
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

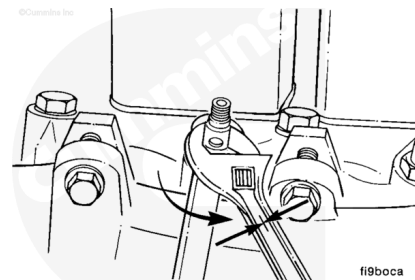
When rust has formed on the hold-down nut, the injector can turn in the bore when the nut is loosened. This can cause severe damage to the head by the injector locating ball cutting a groove in the bore.

Soak the hold-down nut with a rust-penetrating solvent for a minimum of 3 minutes.

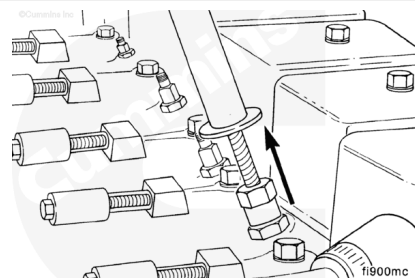
Hit the injector body with the drift pin to loosen any rust.



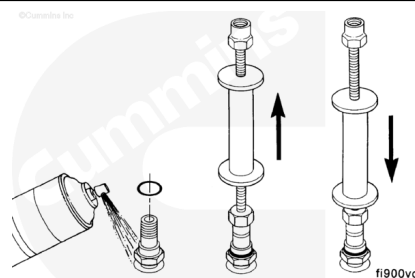
Hold the injector body with an adjustable wrench while loosening the hold-down nut with a 24-mm box wrench.



Use Injector Puller, Part Number 3823276, to remove the injectors. An injector puller is available for difficult-to-remove injectors.

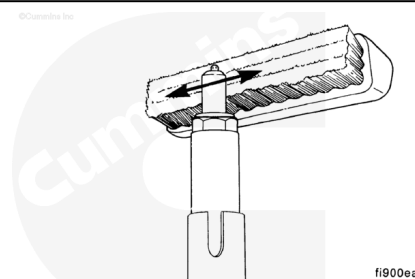


It is often necessary to tap the injector with the injector puller to work the injector up and down to remove it.

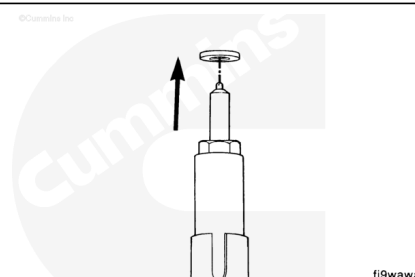


Disassemble

Clean the carbon residue from the injector nozzle. Use a brass wire brush and a piece of hardwood dipped in test oil.

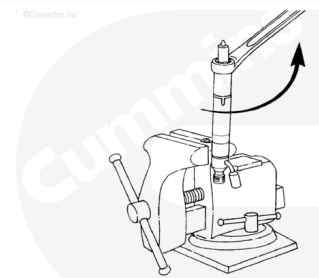


Remove the injector seal and discard.



Clamp an injector hold-down clamp in a soft-jawed vise to hold the injector.

Remove the injector nozzle retaining nut.

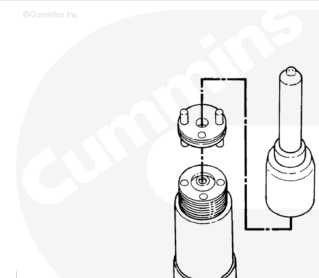


fi9numb

⚠ CAUTION ⚠

Place the injector nozzle and needle valve in a suitable bath of clean test oil or damage will occur.

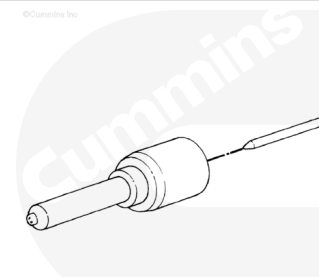
Remove the nozzle needle valve and intermediate plate.



fi9vama

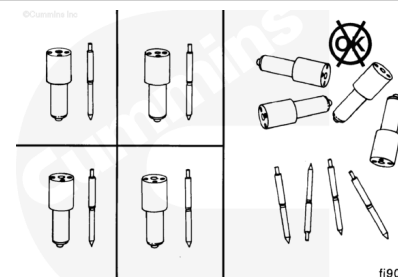
⚠ CAUTION ⚠

Hold the needle valve by the stem only. Contact from the skin will corrode the finely lapped surface.



fi900va

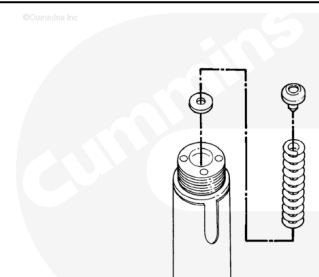
Note : The needle valve and nozzle tip are precisely matched for fit. The parts must not be intermixed.



fi900wa

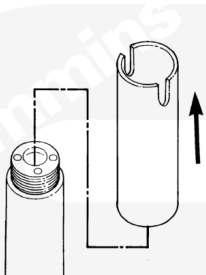
Remove the nozzle holder from the vise.

Remove the pressure spindle, pressure spring, and shims.



fi900fa

Remove and discard the injector sealing sleeve.



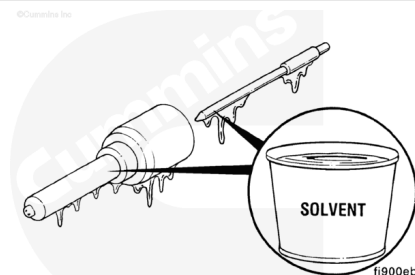
fi9slma

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Rinse the nozzle bodies and needle valves in solvent to flush thoroughly and remove completely all varnish and carbon deposits.

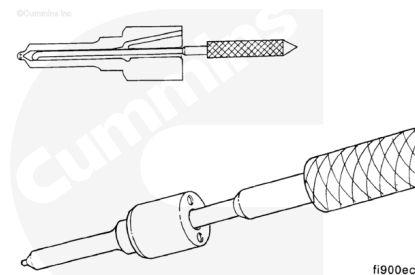


fi900eb

⚠ CAUTION ⚠

Never use emery paper, a steel brush, or any other metal scraper to clean the nozzle. The parts can be damaged.

Dip the nozzle seat in clean test oil, and use the nozzle cleaning kit, Part Number 3376947, to clean the nozzle seat. Polish the needle seat with a piece of hardwood dipped in test oil.

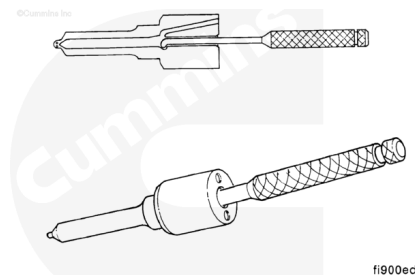


fi900ec

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

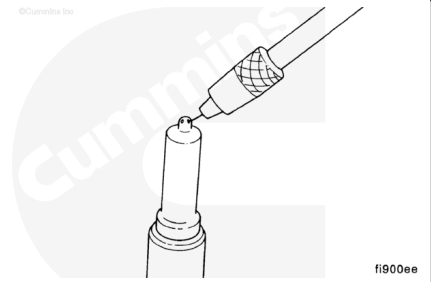
Clean the interior ring groove of the nozzle with a scraper (as shown). Rinse in solvent to remove all dirt and carbon residue, and dip in clean test oil.



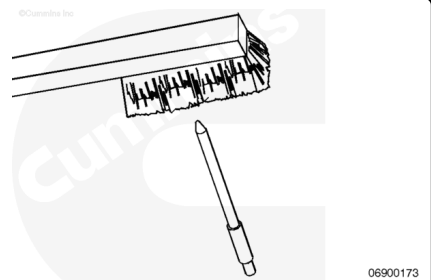
fi900ed

Clean the spray holes (as shown) with an appropriate-size cleaning needle.

Remove burned-on combustion deposits on all nozzles with a commercially available cleaner. Rinse all parts in clean test oil.



Clean the needle valve tip with a brass brush.



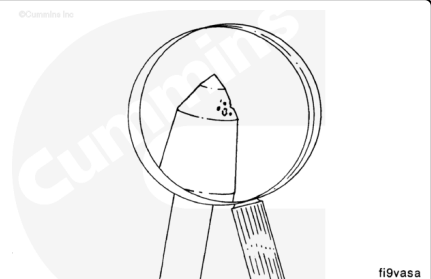
Inspect for Reuse

Inspect the injector. Inspect the o-ring for damage. Inspect for burrs on the inlet to the injector. Check the nozzle holes for any signs of damage such as hole erosion or hole plugging. Also, check the nozzle color for signs of overheating. Overheating will cause the nozzle to turn a dark yellow and tan or blue color, depending on the temperature of the overheat.

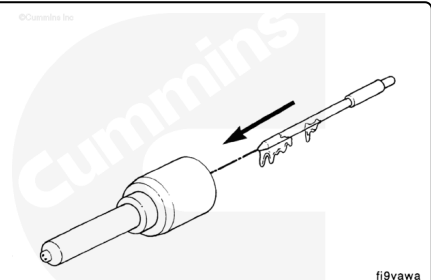
Inspect for rough surfaces and erosion. The pressure shoulder will normally have a rough machined appearance.

Inspect the injector bore for old sealing washers.

Note : Deteriorated needle valves must be replaced as a matched unit with their compatible nozzle body.



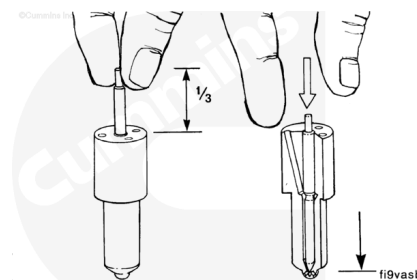
Dip the needle valve in clean test oil, and insert the needle valve all the way into the nozzle body.



Pull the needle valve one-third of the way out of the nozzle body. With the needle valve in the vertical position, the needle valve **must** slide all the way back into the nozzle body under its own weight.

If the nozzle fails the slide test, clean and test the nozzle again.

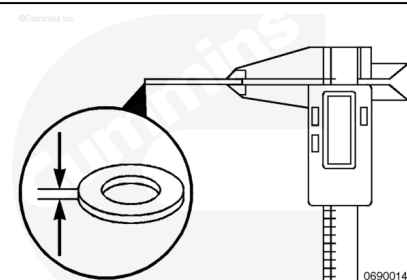
Note : Any needle valve and nozzle body assembly that does not pass this test must be replaced.



Measure

Verify that the injector sealing washer is the correct thickness. The incorrect sealing washer can cause high-pressure fuel leaks and/or performance problems due to incorrect injector protrusion.

Shim thickness specification: 1.5 mm [0.06 in].



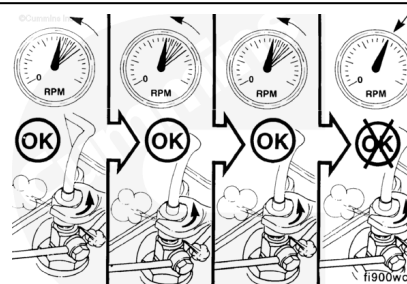
Test

To find which cylinder is misfiring, operate the engine and loosen the fuel line nut at one injector, and listen for a change in engine speed.

Note : A drop in engine speed indicates the injector was delivering fuel to the cylinder.

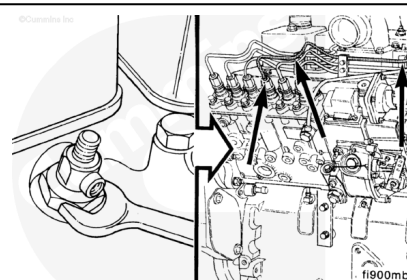
Note : Be sure to tighten the fuel line nut before proceeding to the next injector.

Check each cylinder until the malfunctioning injector is found.

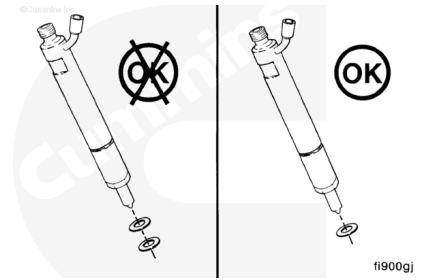


Remove the malfunctioning injector to test or replace it.

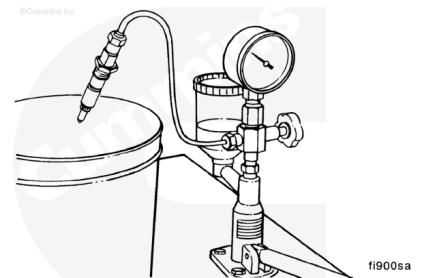
If the engine continues to misfire after replacing the injector, check for leaks in the high-pressure line. Also, check for a defective delivery valve that lets the fuel drain back into the injection pump.



Check for an extra copper sealing washer on the injector.

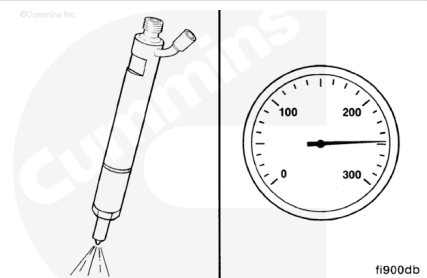


Carbon buildup in the orifices in the nozzle will also cause low power from the engine. Remove and check the spray pattern, or replace the injectors.



⚠ WARNING ⚠

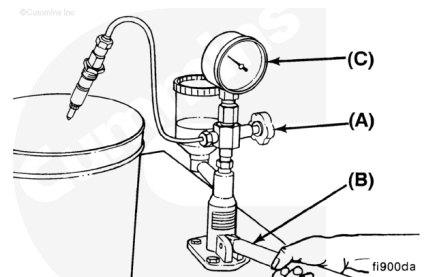
While testing the injectors, keep hands and body parts away from the injector nozzle. Fuel coming from the injector is under extreme pressure and can cause serious injury by penetrating the skin.



Note : Using Injector Nozzle Tester, Part Number 3376947, all nozzles must be tested for opening pressure, chatter, and spray pattern.

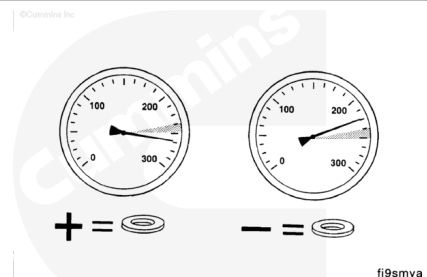
Check the injector opening pressure.

1. Open the valve.
2. Operate the lever at one stroke every second.
3. Read the pressure indicated when the injector spray begins.



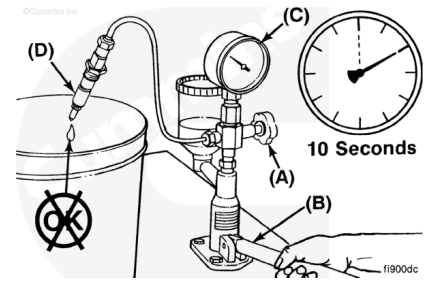
If the opening pressure does **not** meet specifications, try one of the following solutions:

1. Add shims to increase pressure.
2. Remove shims to decrease pressure.



Leakage Test

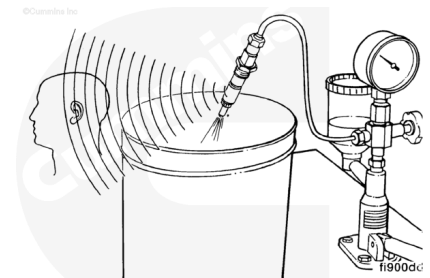
1. Open the valve (A).
2. Operate the lever (B) to maintain a pressure of 20 bar [290 psi] below opening pressure (C).
3. No drops should fall from the tip (D) within 10 seconds.



Chatter Test

The chatter test indicates the ability of the needle valve to move freely and atomize the fuel correctly. An audible sound will possibly be heard as the valve rapidly opens and closes. A well-optimized spray pattern will possibly be seen.

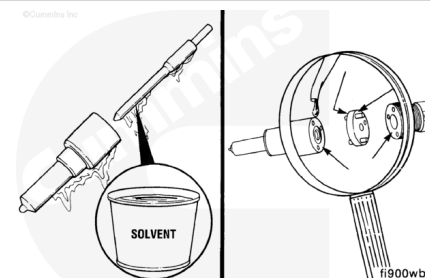
Note : Used nozzles **must not** be evaluated for chatter at lower speeds. A used nozzle can generally be used if it passes the leakage test.



Assemble

⚠ WARNING ⚠

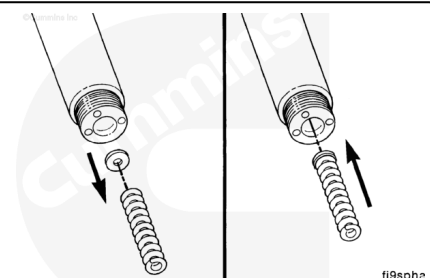
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



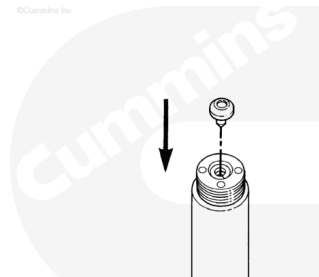
Note : Make sure that all mating surfaces and pressure faces are thoroughly cleaned and lubricated with test oil before assembly. New nozzles must be cleaned and lubricated before assembly.

Note : Install the same thickness of shims that were removed in disassembly. Use the pressure spring to make sure that the shims are installed flat.

Install the shims and pressure spring.

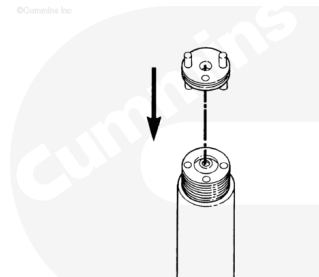


Install the spindle.



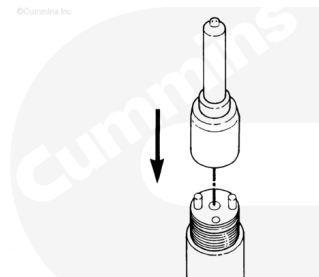
fi900aa

Install the intermediate plate.



fi9plaa

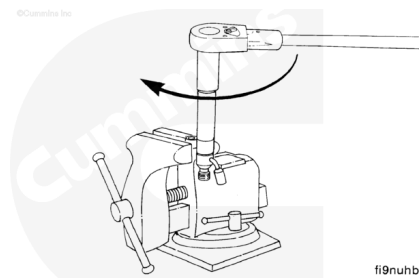
Install the needle valve and nozzle assembly.



fi9vaaa

Install the nozzle retaining nut.

Torque Value: 30 n·m [22 ft-lb]



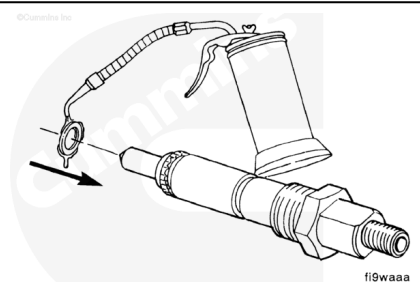
fi9nuhb

Install

Assemble the injector and new copper sealing washer.

Use **only** one copper washer.

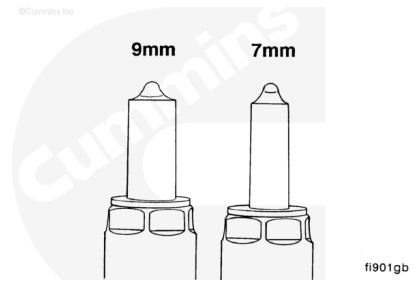
Service Tip: A light coat of clean lubricating engine oil between the washer and injector can help to keep the washer from falling during installation.



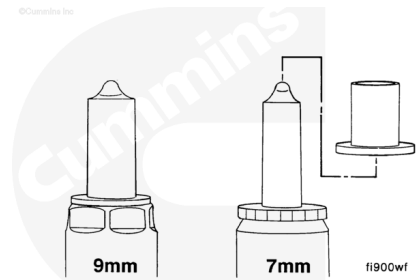
fi9waaa

⚠ CAUTION ⚠

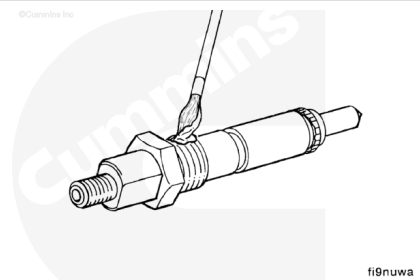
Early model injectors (pre-1991) have a 9-mm injector tip that can not be used in engines built in 1991 or later as these engines use a 7-mm injector tip.



Seven mm injectors can be used in early model (9-mm) injector holes provided the special adapter sleeve is installed onto the 7-mm injector tip.



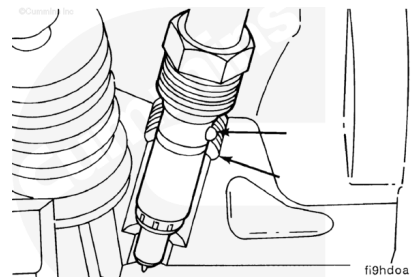
Apply a coat of anti-seize compound, Part Number 3824879, to the threads of the injector hold-down nut and between the top of the nut and the injector body.



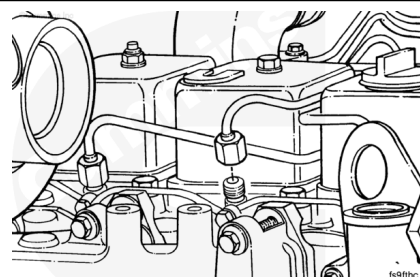
Note : Align the injector's protrusion with the notch in the bore.

Note : The present Bosch® injector has an o-ring located above the hold down nut. After tightening the injector, be sure to push the o-ring into the groove.

Torque Value: 60 n•m [44 ft-lb]

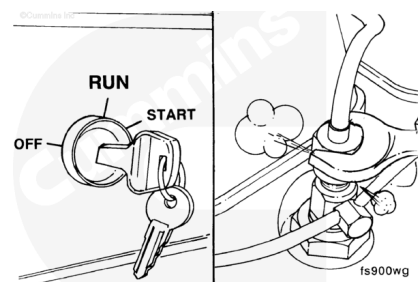


Assemble the fuel drain manifold and high-pressure lines. Leave the high-pressure fittings loose at the injectors.



⚠ WARNING ⚠

It is necessary to turn the keyswitch to the ON position. Because the engine can start, be sure to follow all of the safety precautions. Use the normal engine starting procedure.



Crank the engine to allow entrapped air to bleed from the lines.

Tighten the fittings.

Torque Value: 38 n•m [28 ft-lb]